

Hashing



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Hashing

Hashing in the data structure is a technique that is used to quickly identify a specific value within a given array.

Hashing is used to store and retrieve an element in $O(1)$ time.

For example:

Search key : (24, 52, 91, 65, 48, 83)

Hash function: $\text{key} \% 10$

Now create a hash table using the search key and the hash function.

Hashing

$24 \% 10 = 4$; so, 24 will be stored at the index 4.
 $52 \% 10 = 2$; so, 52 will be stored at the index 2.
 $91 \% 10 = 1$; so, 91 will be stored at the index 1.
 $65 \% 10 = 5$; so, 65 will be stored at the index 5.
 $48 \% 10 = 8$; so, 48 will be stored at the index 8.
 $83 \% 10 = 3$; so, 83 will be stored at the index 3.

Index	Value
0	
1	91
2	52
3	83
4	24
5	65
6	
7	
8	48
9	

Hash Table

Hashing

If the new elements whose remainder is 5 (like 75) are added in the above hash table, then 75 will be added next to 65 and the table will be as like as below:

Index	Value
0	
1	91
2	52
3	83
4	24
5	65, 75, ...
6	
7	
8	48
9	

Hash Table

If added more elements

Collision

Hashing

At an index position, two or more elements cannot be placed. This is a problem called Collision Resolution which is solved by using open hashing or close hashing.

For open hashing

A Link list is added to the previous index location like below.

}

Inde x	Value
0	
1	91
2	52
3	83
4	24
5	65
6	
7	
8	48
9	

→ 75

Hash Table

Thank You